Quiz 4: Combining Logical Functions II: Nested IFS

Date: 13/1/2026.

1. For the arguments to the **IF** function, other **IF** functions may be embedded within them.

   1 / 1 point

   Of the 3 arguments to the **IF** function, which can have another **IF** nested in them?

   - [x] First argument: **logical_test**

   Correct
   Yes, the **logical_test** argument can contain a nested **IF** function.

   - [x] Second argument: **value_if_true**

   Correct
   Yes, the **value_if_true** argument can contain a nested **IF** function.

   - [x] Third argument: **value_if_false**

   Correct
   Yes, the **value_if_false** argument can contain a nested **IF** function.

2. To detect whether the cells **X1**, **Y1**, and **Z1** contain **exactly 2 values** which are the same (e.g. **X1** and **Z1** are the same and **Y1** is different), the coding for giving an output "Duplicate" could look like:

   1 / 1 point

   - (●) =IF(X1=Y1,IF(X1=Z1,"","Duplicate"),IF(OR(X1=Z1,Y1=Z1),"Duplicate",""))
   - ( ) =IF(X1<>Y1,IF(X1=Z1,"Duplicate",""),"")
   - ( ) =IF(X1=Y1,IF(X1=Z1,"","Duplicate"),IF(AND(X1=Z1,Y1=Z1),"Duplicate",""))

   Correct
   Correct, the first input is a logical test and the other 2 are the outcomes that depend on whether the test is true or false.

3. To detect whether the values in cells **X1**, **Y1**, and **Z1** are in descending order (with no ties), the coding for giving an output "Descending Order" could look like:

   1 / 1 point

   - ( ) =IF(X1>Y1,IF(X1<=Z1,"","Descending Order"),"")
   - ( ) =IF(X1>Z1,IF(Y1<=Z1,"","Descending Order"),"")
   - ( ) =IF(Y1>Z1,IF(X1<=Z1,"","Descending Order"),"")
   - (●) =IF(X1>Y1,IF(Y1>Z1,"Descending Order",""),"")

   Correct
   Correct, the first test in the "outer" IF must hold as well as the second test in the "inner" IF.

4. Consider the code **=IF(X1>=Y1,"", IF(Y1>=Z1,"","tick"))**.

1 / 1 point

Supposing the output "tick" is produced, which of the following could be true about **X1**, **Y1**, **Z1**?

Please note that ties are allowed for this question.

- (selected) **X1**, **Y1**, **Z1** are in ascending order.
- **X1**, **Y1**, **Z1** are in descending order.
- **X1** is greater than **Y1** and **Y1** equals **Z1**.
- **X1** and **Y1** are equal and **Z1** is less than **X1**.

Correct
Correct response. This scenario could be true about **X1**, **Y1**, **Z1**.